

1 Normalization: 3NF/BCNF Proof

In this section we formally justify that each relation in our schema is in Boyce–Codd Normal Form (BCNF), and therefore also in Third Normal Form (3NF).

Relations and Functional Dependencies

Below we list, for each relation R , its attributes, primary key(s), and the set of non-trivial FDs we assume for the cleaned dataset.

`movies`

$R_{\text{movies}}(\text{movie_id}, \text{title}, \text{release_year}, \text{runtime_minutes}, \text{is_adult}, \text{rating}, \text{num_votes})$

- Primary key: `{movie_id}`.

- Non-trivial FDs:

$\text{movie_id} \rightarrow \{\text{title}, \text{release_year}, \text{runtime_minutes}, \text{is_adult}, \text{rating}, \text{num_votes}\}.$

`genres`

$R_{\text{genres}}(\text{genre_id}, \text{genre_name})$

- Candidate keys: `{genre_id}` and `{genre_name}` (since `genre_name` is unique).

- Non-trivial FDs:

$\text{genre_id} \rightarrow \text{genre_name}, \quad \text{genre_name} \rightarrow \text{genre_id}.$

`movie_genres`

$R_{\text{movie_genres}}(\text{movie_id}, \text{genre_id})$

- Primary key: `{movie_id, genre_id}`.

- Non-trivial FDs: the only non-trivial FD is

$(\text{movie_id}, \text{genre_id}) \rightarrow (\text{all attributes}).$

`people`

$R_{\text{people}}(\text{person_id}, \text{name}, \text{birth_year}, \text{death_year})$

- Primary key: `{person_id}`.

- Non-trivial FDs:

$\text{person_id} \rightarrow \{\text{name}, \text{birth_year}, \text{death_year}\}.$

roles

$R_{\text{roles}}(\text{movie_id}, \text{person_id}, \text{role})$

- Primary key: $\{\text{movie_id}, \text{person_id}, \text{role}\}$.
- Non-trivial FDs:

$(\text{movie_id}, \text{person_id}, \text{role}) \rightarrow (\text{all attributes}).$

person_known_for

$R_{\text{person_known_for}}(\text{person_id}, \text{movie_id})$

- Primary key: $\{\text{person_id}, \text{movie_id}\}$.
- Non-trivial FDs:

$(\text{person_id}, \text{movie_id}) \rightarrow (\text{all attributes}).$

market_info

$R_{\text{market_info}}(\text{tmdb_id}, \text{imdb_id}, \text{budget}, \text{revenue}, \text{popularity}, \text{overview}, \text{poster_path})$

- Primary key: $\{\text{tmdb_id}\}$.
- In the cleaned dataset, there is at most one row per `tmdb_id` and at most one row per `imdb_id`. The main FDs are:

$\text{tmdb_id} \rightarrow \{\text{imdb_id}, \text{budget}, \text{revenue}, \text{popularity}, \text{overview}, \text{poster_path}\},$

and (conceptually) also

$\text{imdb_id} \rightarrow \{\text{tmdb_id}, \text{budget}, \text{revenue}, \text{popularity}, \text{overview}, \text{poster_path}\}.$

Thus both `tmdb_id` and `imdb_id` behave as candidate keys.

countries

$R_{\text{countries}}(\text{country_name})$

- Primary key: $\{\text{country_name}\}$.
- There are no non-trivial FDs other than the key determining itself.

movie_production_countries

$R_{\text{movie_production_countries}}(\text{tmdb_id}, \text{country_name})$

- Primary key: $\{\text{tmdb_id}, \text{country_name}\}$.
- Non-trivial FDs:

$(\text{tmdb_id}, \text{country_name}) \rightarrow (\text{all attributes}).$

BCNF Verification

We now check the BCNF condition relation by relation.

- **movies:** The only non-trivial FD has determinant $\{\text{movie_id}\}$, which is the primary key. Hence every non-trivial FD has a key on the left-hand side, so **movies** is in BCNF.
- **genres:** The determinants of all non-trivial FDs are $\{\text{genre_id}\}$ or $\{\text{genre_name}\}$, both of which are candidate keys. Therefore **genres** is in BCNF.
- **movie_genres:** The only non-trivial FD has determinant $\{\text{movie_id}, \text{genre_id}\}$, which is the primary key. Thus **movie_genres** is in BCNF.
- **people:** The determinant of the only non-trivial FD is the primary key $\{\text{person_id}\}$, so **people** is in BCNF.
- **roles:** The only non-trivial FD is from the composite key $\{\text{movie_id}, \text{person_id}, \text{role}\}$ to all attributes. Hence **roles** is in BCNF.
- **person_known_for:** The only non-trivial FD is from the composite key $\{\text{person_id}, \text{movie_id}\}$ to all attributes. Therefore **person_known_for** is in BCNF.
- **market_info:** Every non-trivial FD has determinant either $\{\text{tmdb_id}\}$ or $\{\text{imdb_id}\}$, each of which is a candidate key in the cleaned dataset. Thus **market_info** is in BCNF.
- **countries:** The only key is $\{\text{country_name}\}$, and there are no non-trivial FDs with a non-key determinant. Therefore **countries** is in BCNF.
- **movie_production_countries:** The only non-trivial FD is from the composite key $\{\text{tmdb_id}, \text{country_name}\}$ to the full tuple. Hence **movie_production_countries** is in BCNF.

Since every relation in our schema satisfies the BCNF condition, the entire schema is in BCNF and therefore also in 3NF, meeting the normalization requirement of the project.